Heat Islands with Several Boxes

Student sheet

Name: _____ **Date:** _____

Safety: Never leave a box unattended, and keep away from cars. The box is not waterproof: if it rains, we go in.

1. What I see on the screen

The **Air temperature** chart has one **coloured line per box**, with its name. Below it, the ΔT card shows how much the warmest box differs from the coolest; the list on the right gives each box's temperature. All of them are **averages over the last minute**. The **Record the scatter** button is pressed with the boxes side by side. You save with **Export data**, top right. **There is no green button here, no Back, no Finish.**

The app's own verdict compares ΔT with the scatter the button recorded; without a press, it assumes 1 °C. **I compare with the scatter written on my sheet.** Above 3 °C, I first check the **white screens**: sun on a casing makes false differences. I wait **five minutes** at each location and I do not hold the box in my hand.

2. What I do on my own

Weather: _____ **Height used:** _____ m **How many boxes are in the list?** _____

First I predict. I do not change the prediction after I measure. 1. The warmest location will be _____, the coolest _____. **2.** The difference between them will be _____ °C. **3.** Two boxes placed *side by side* will show numbers that differ by _____ °C.

Co-location. All boxes side by side, on the same table, out of the sun, **five minutes**. Only then do I read and press **Record the scatter**.

#	Box name, as shown on screen	T (°C)	Where I take it
1			
2			
3			
4			

SCATTER = largest − smallest = _____ − _____ = _____ °C

Now I measure in the field. The same height at all boxes, a white screen on *every* box, five minutes, the time noted. Two compared boxes differ in a single thing: the surface *or* the sun.

Location (surface)	Box	Time	T (°C)	RH (%)

ΔT = warmest _____ − coolest _____ = _____ °C Warmest box: _____

If I repeat the route (project route), I record for each outing the time, the scatter and ΔT. **The scatter is measured again every time.**

Morning: _____ / _____ / _____ Noon: _____ / _____ / _____ End of day: _____ / _____ / _____

3. The conclusion I write

My ΔT: _____ °C. Our scatter: _____ °C. So the difference ☐ is detectable ☐ **is not detectable** with our setup. *"Not detectable" does not mean "does not exist".*

I predicted that, and the data show that _____

_____

Our scatter was _____

_____

The warmest location was the warmest because _____

_____

Exit ticket

One thing that surprised you today, and one thing you would do differently if you repeated the measurement.

_____